STAT 387 (Introduction to Statistical Learning, Winter 2026)
Project 8

Instructions:

- Due date: March 16
- Submit a typed report and slideshow presentation.
- Do a good job.

Preamble:

Consider the Bank Marketing data available in the **`bank_marketing.csv`**. A description of the variables is given in the file **`bank_marketing_readme.txt`**. You can read more about the data on https://archive.ics.uci.edu/dataset/222/bank+marketing. The data is related with direct marketing campaigns (phone calls) of a Portuguese banking institution. The primary goal is to predict if the client will subscribe a term deposit (variable y) based on other inputs.

Randomly split the dataset into 80% training and 20% as test.

Problem statements:

a) Perform an exploratory analysis of data.

b) Fit a KNN with K chosen optimally using training error rate. Report both the training and test MSE rates for the optimal K.

c) Now report test MSE using Random Forest with B = 500, and B = 1000. Compare your findings.

d) Repeat (c) with a Boosting approach with B = 1000, d = 1, and $\lambda = 0.01$.

e) Repeat (c) with an SVM approach based on a radial basis function (choose any γ) and a dth degree polynomial kernel (use d = 2 and d = 3).

f) Compare the results from the various methods above. Which method would you recommend? Justify your answer.